IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

Vsevolod Muronets
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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Goal: Predict whether the Falcon 9 first stage will land successfully.

Methodology summary:

- Data collected via SpaceX REST API and Wikipedia web scraping
- Data cleaning, feature engineering (One-Hot Encoding)
- EDA via visualization (matplotlib/seaborn) and SQL queries
- Interactive Folium map and Plotly Dash dashboard
- Four classification models trained and compared.

Key results:

- Overall landing success rate: 66.7%
- Success rate grows year-over-year (2013–2020)
- Best model by CV-score: Decision Tree (87.7%)
- All 4 predictive models achieved test accuracy: 83.3%

Introduction

Project context:

SpaceX offers Falcon 9 launches at $62M - far below competitors ($165M+). The key advantage is first stage reusability. Predicting a successful landing allows estimation of the true launch cost - valuable for any company seeking to compete with SpaceX.

Research questions:

- What factors influence first stage landing success?
- How do launch site, orbit type, and payload mass affect outcomes?
- Which machine learning model best predicts the result?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - GET requests to SpaceX REST API (/v4/launches/past) + Wikipedia web scraping with BeautifulSoup
- Perform data wrangling
  - Filtered Falcon 9 only, imputed missing PayloadMass values with the mean, created binary target variable Class
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - 4 classification models (logistic regression, support vector machines, decision tree, k-nearest neighbors), GridSearchCV with cv=10, StandardScaler normalization.

Data Collection

Two independent data sources were used:

Source 1 - **SpaceX REST API:**

- Request to https://api.spacexdata.com/v4/launches/past
- Extracted: booster version, payload mass, orbit, launch site (with coordinates), landing outcome
- Filtered to Falcon 9 only (Falcon 1 removed)
- Final dataset: 90 records, 17 features

Source 2 - **Wikipedia Web Scraping:**

- URL: Wikipedia "List of Falcon 9 and Falcon Heavy launches" (snapshot: June 9, 2021)
- Tools: requests + BeautifulSoup
- Extracted columns: Flight No., Launch site, Payload, Payload mass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date

Data Collection – SpaceX API

Process:

GET request to /v4/launches/past -> status 200
JSON normalized via pd.json_normalize()
Four helper functions to enrich data:

getBoosterVersion() - booster name from /v4/rockets/
getLaunchSite() - site name and coordinates from /v4/launchpads/
getPayloadData() - mass and orbit from /v4/payloads/
getCoreData() - block, reuse count, serial, landing outcome from /v4/cores/

Filtered to Falcon 9 only (4 Falcon 1 records removed).
Missing values: PayloadMass (5 values) -> replaced with mean; LandingPad (26 values) -> kept as None.

- GitHub URL: https://github.com/Vsevolod-Muronets/Porfolio/blob/9269cc49a2c9b0fe85bda546c25cbea2f3765534/IBM_Data_Science_Professional_Certificate/Applied%20Data%20Science%20Capstone%20(Final%20Project)/Module%201%20%20-%20Collecting%20the%20Data%20and%20Data%20Wrangling/jupyter-labs-spacex-data-collection-api.ipynb

SpaceX Rest API

↓ (GET /v4/launches/past)

json_normalize()

↓

Helper API calls:

getBoosterVersion - getLaunchSite - getPayloadData - getCoreData

↓

Filter: Falcon 9 only

Remove Falcon 1 records

↓

Handle missing values

PayloadMass -> mean; LandingPad -> None

Data Collection - Scraping

Process:

1. HTTP GET to a static Wikipedia snapshot (June 9, 2021)
2. BeautifulSoup parses HTML, finds `<table class="wikitable">`
3. Column names extracted from `<th>` headers: `['Flight No.', 'Launch site', 'Payload', 'Payload mass', 'Orbit', 'Customer', 'Launch outcome', 'Version Booster', 'Booster landing', 'Date']`
4. All `<tr>` rows iterated across all launch tables -> dictionary populated
5. Converted to DataFrame -> `spacex_web_scraped.csv`

Challenges: Wikipedia contains annotations (`B0004.1[8]`), missing values (`N/A [e]`), inconsistent formatting - handled via custom helper functions (`date_time()`, `booster_version()`, `landing_status()`, etc.)

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Wikipedia URL
↓
requests.get()
↓
BeautifulSoup
↓
find_all('table', 'wikitable')
↓
extract columns from <th>
↓
parse <tr> rows
↓
DataFrame

Data Wrangling

Dataset overview:

90 Falcon 9 records, 17 features

Launch site distribution: CCAFS SLC 40 - 55, KSC LC 39A - 22, VAFB SLC 4E - 13

Top orbits: ISS - 21, VLEO - 14, PO - 9, LEO - 7, SSO - 5

Missing values: LandingPad - 28.9% (kept as None - not all launches attempted a pad landing).

Landing outcomes (top-5):

Outcome	Count
True ASDS (drone ship, success)	41
None None (no attempt)	19
True RTLS (ground pad, success)	14
False ASDS (drone ship, failure)	6
True Ocean (ocean, success)	5

Target variable Class:

Class = 1 - successful landing (True ASDS, True RTLS, True Ocean)

Class = 0 - unsuccessful or no landing attempt

Overall success rate: 66.7%

GitHub URL:
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dataset_part_1.csv

↓

Null analysis

↓

value_counts() by site/orbit/outcome

↓

Define bad_outcomes set

↓

Binary Class variable

EDA with Data Visualization

Seven chart types built with seaborn and matplotlib were plotted:

1. FlightNumber vs PayloadMass - catplot with hue=Class: as flight number increases, landing success improves; even heavy payloads increasingly land successfully.

2. FlightNumber vs LaunchSite - scatter: CCAFS SLC 40 has the most launches; KSC LC 39A shows high success in later flights.

3. PayloadMass vs LaunchSite - scatter: VAFB SLC 4E has no launches above 10,000 kg.

4. Success Rate by Orbit - bar chart: GEO, HEO, SSO, ES-L1 show 100% success (small sample); GTO is the most inconsistent.

5. FlightNumber vs Orbit - scatter: LEO success correlates with flight number; GTO shows no such trend.

6. PayloadMass vs Orbit - scatter: with heavy payloads the successful landing or positive landing rates are more for PO,LEO and ISS. For GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

7. Yearly Trend (2010–2020) - line chart: success rate steadily increases from 2013 to 2020.

GitHub URL:

https://github.com/Vsevolod-Muronets/Porfolio/blob/9269cc49a2c9b0fe85bda546c25cbea2f3765534/IBM_Data_Science_Professional_Certificate/Applied%20Data%20Science%20Capstone%20(Final%20Project)/Module%202%20-%20EDA%20using%20SQL%2C%20Pandas%2C%20Seaborn%20and%20Matplotlib/edadataviz.ipynb

EDA with SQL

10 SQL queries were executed against a SQLite database (my_data1.db) loaded from Spacex.csv (101 records):

Task 1: SELECT DISTINCT Launch_Site -> 4 unique sites
Task 2: WHERE Launch_Site LIKE ‘CCA%’ LIMIT 5 -> 5 records from Cape Canaveral
Task 3: %sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = ‘NASA (CRS)’ -> 45 596 kg. total payload
Task 4: AVG(PAYLOAD_MASS__KG_) WHERE Booster_Version LIKE ‘F9 v1.1%’ -> 2,534.7 kg
Task 5: First successful ground landing -> 2015-12-22
Task 6: Drone ship success with payload 4 000-6 000 kg -> 4 boosters (F9 FT series)
Task 7: Mission outcomes grouped -> 100 successes, 1 failure
Task 8: Boosters with max payload -> 12 F9 B5 boosters
Task 9: 2015 drone ship failures -> January (B1012) and April (B1015), both from CCAFS LC-40
Task 10: Landing outcomes ranked 2010–2017 -> “No attempt” leads with 10, drone ship 5 successes / 5 failures

GitHub URL:
https://github.com/Vsevolod-Muronets/Porfolio/blob/9269cc49a2c9b0fe85bda546c25cbea2f3765534/IBM_Data_Science_Professional_Certificate/Applied%20Data%20Science%20Capstone%20(Final%20Project)/Module%202%20-%20EDA%20using%20SQL%2C%20Pandas%2C%20Seaborn%20and%20Matplotlib/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

I have built an interactive map using the Folium library with three layers of analysis:

Task 1 - All launch sites marked: folium.Circle + folium.Marker with DivIcon text labels placed at each site's coordinates.

Task 2 - Launch outcomes color-coded: MarkerCluster plugin added with green/red markers per launch record. New column marker_color derived from class: 1 → green, 0 → red.

Task 3 - Proximity distances calculated: MousePosition plugin enabled coordinate tracking. Distance formula (calculate_distance()) was used to measure distances from CCAFS LC-40 to coastline, railway, highway (all < 1 km), and nearest city (~23 km). folium.PolyLine objects were drawn between launch site and each proximity point with distance labels via DivIcon.

Key findings:

- All sites are coastal and close to transport infrastructure;
- Sites are kept far from populated areas for safety;
- Florida sites are positioned as far south as US territory allows.

GitHub URL:

https://github.com/Vsevolod-Muronets/Porfolio/blob/9269cc49a2c9b0fe85bda546c25cbea2f3765534/IBM_Data_Science_Professional_Certificate/Applied%20Data%20Science%20Capstone%20(Final%20Project)/Module%203%20-%20Interactive%20Visual%20Analytics%20and%20Dashboard/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Interactive dashboard built with Plotly Dash featuring two plots and two interactive controls:

Controls:

- dcc.Dropdown - select ALL sites or a specific launch site
- dcc.RangeSlider - filter by payload mass (0-10 000 kg, step 1 000)

Plot 1 - Pie chart (success-pie-chart):

- ALL selected -> pie chart showing total successful launches per site (values='class', names='Launch Site')
- Site selected -> Success vs. Failed breakdown for that site (names='class')
- Callback: Input('site-dropdown') -> Output('success-pie-chart')

Plot 2 - Scatter plot (success-payload-scatter-chart):

- X = Payload Mass (kg), Y = class (0/1), color = Booster Version Category
- Filtered by both payload range (RangeSlider) and selected site (Dropdown)
- Callback: Input('site-dropdown') + Input('payload-slider') -> Output('success-payload-scatter-chart')

GitHub URL:

https://github.com/Vsevolod-Muronets/Portfolio/blob/9269cc49a2c9b0fe85bda546c25cbea2f3765534/IBM_Data_Science_Professional_Certificate/Applied%20Data%20Science%20Capstone%20(Final%20Project)/Module%203%20-%20Interactive%20Visual%20Analytics%20and%20Dashboard/spacex-dash-app.py

Predictive Analysis (Classification)

Data preparation:

1. Target variable Y: Class column converted to NumPy array (90 values: 0 or 1);
2. Features X: One-Hot encoded dataset (dataset_part_3.csv);
3. Standardization: StandardScaler applied to all features;
4. Split: 80/20 train/test, random_state=2 -> 72 training, 18 test samples.

Model development pipeline (GridSearchCV, cv=10):

Model	**Parameters searched**
Logistic Regression	C: [0.01, 0.1, 1], penalty: l2
SVM	kernel: linear/rbf/poly/sigmoid, C: logspace(-3,3,5), gamma: logspace(-3,3,5)
Decision Tree	criterion, splitter, max_depth, max_features, min_samples_leaf, min_samples_split
KNN	n_neighbors: 1–10, algorithm: auto/ball_tree/kd_tree/brute, p: [1,2]

Best hyperparameters found:

- LR: 'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs';
- SVM: 'C': 1.0, 'gamma': ~0.03, 'kernel': 'sigmoid;
- DT: 'criterion': 'entropy', 'max_depth': 4, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split': 2, 'splitter': 'random';
- KNN: 'algorithm': 'auto', 'n_neighbors': 10, 'p': 1

GitHub URL:

https://github.com/Vsevolod-Muronets/Portfolio/blob/9269cc49a2c9b0fe85bda546c25cbea2f3765534/IBM_Data_Science_Professional_Certificate/Applied%20Data%20Science%20Capstone%20(Final%20Project)/Module%204%20-%20Predictive%20Analysis%20(Classification)/SpaceX_Machine_Learning_Prediction_Part_5.ipynb

dataset_part_2.csv + dataset_part_3.csv to df

↓

StandardScaler

↓

train_test_split (80/20)

↓

GridSearchCV (cv=10)

↓

4 models trained

↓

Compare test and CV-scores

↓

Best model selected

Results

Exploratory data analysis results:

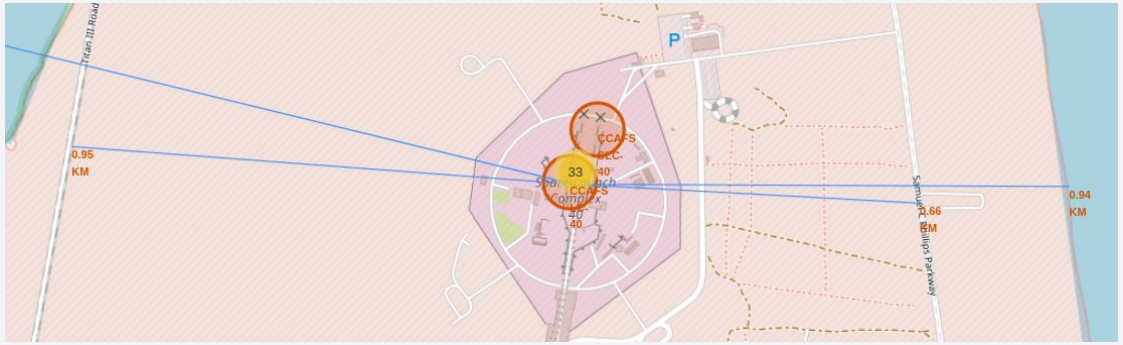
- 90 Falcon 9 launches analyzed (2010–2020)
- Overall landing success rate: 66.7%
- Most active site: CCAFS SLC 40 (55 launches)
- Success rate grew from ~0% in 2010 to near 100% by 2020
- Orbits with 100% success: GEO, HEO, SSO, ES-L1

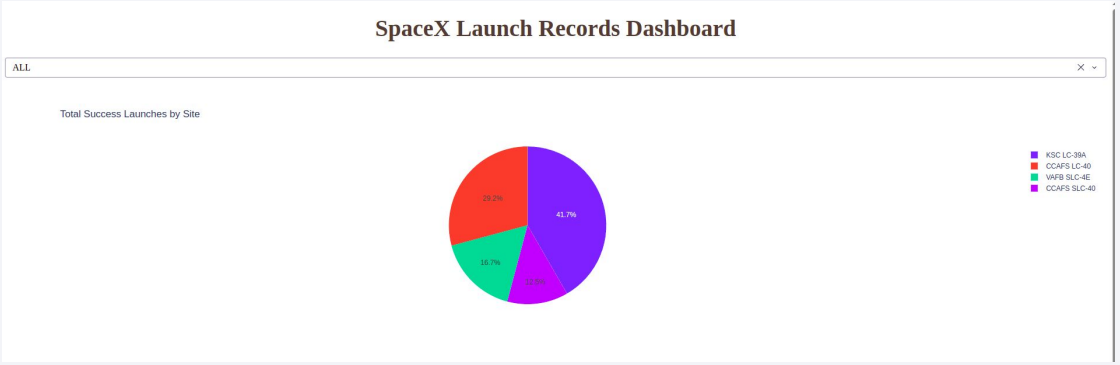
Interactive analytics highlights:

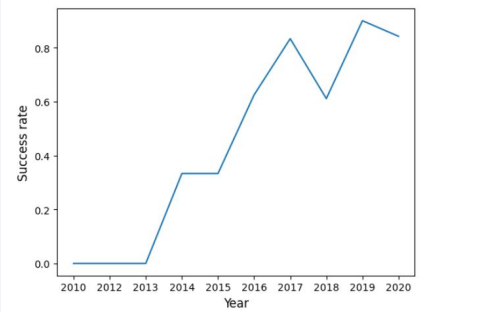
- All 4 launch sites located within 1 km of coastline
- KSC LC 39A has the highest success ratio among all sites
- B5 boosters dominate high-payload successful landings

Predictive analysis results:

- Best CV accuracy: Decision Tree at 87.7%
- All models converged to 83.3% test accuracy (18-sample test set)
- Main error type: False Positives (predicted landing when there was none)



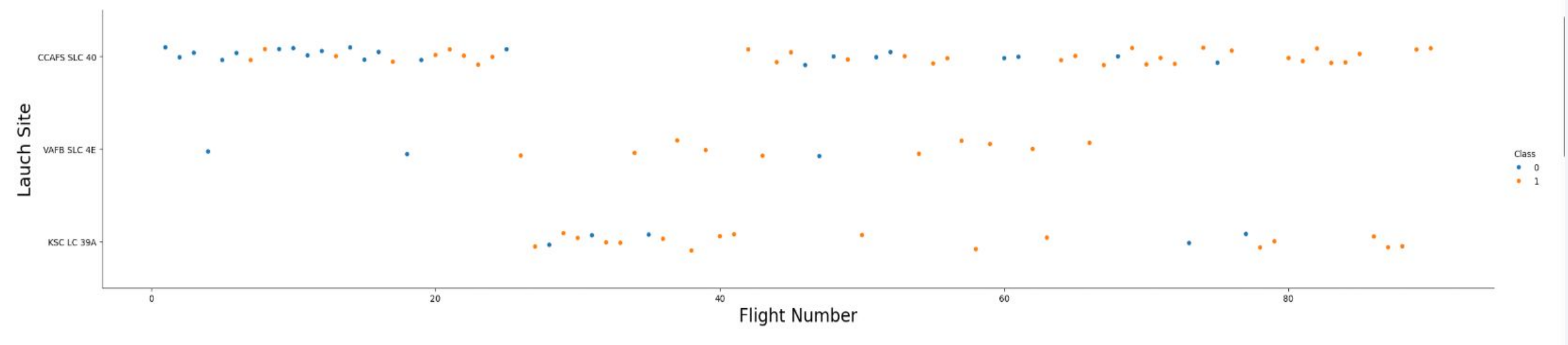




Section 2

Insights drawn from EDA

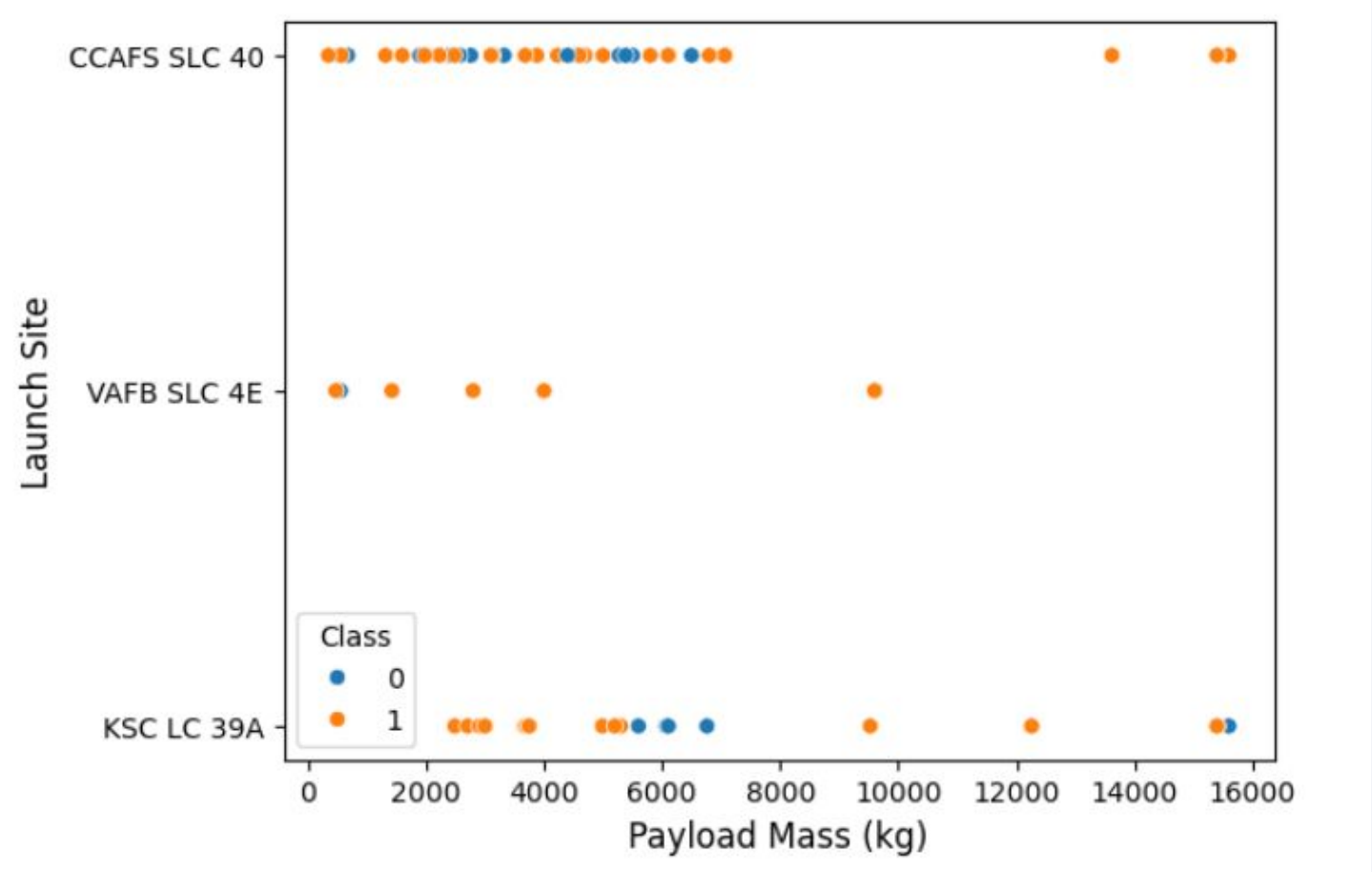
Flight Number vs. Launch Site



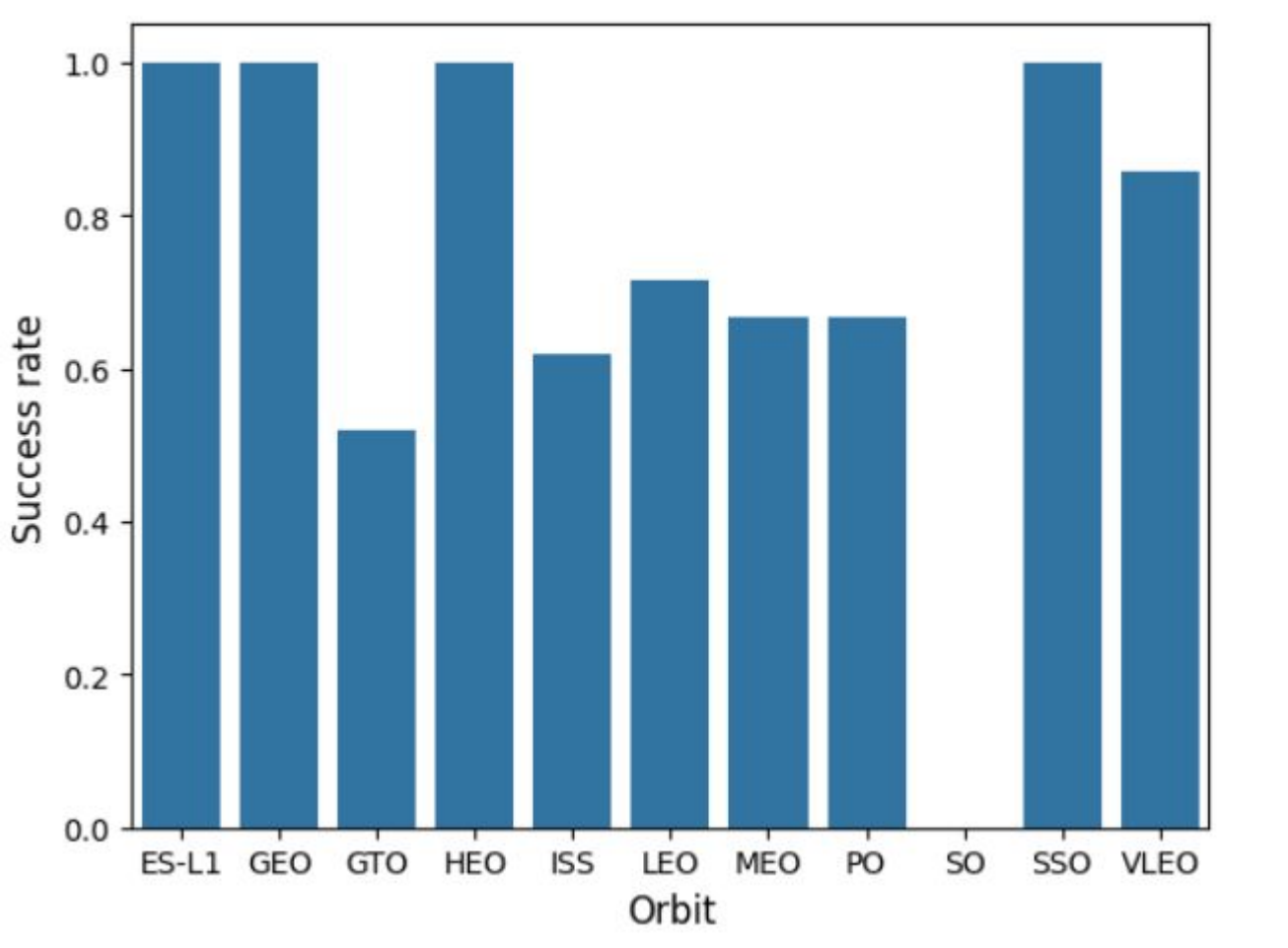
CCAFS SLC 40 is the most active and eldest site with 55 launches, while other sites have higher success rates. Early launches are almost all unsuccessful in terms of landing (Class=0). As flight number increases, the share of successful landings grows across all sites, what means that with time the success rate grows. All of the sites show rather high success rates in later missions.

Payload vs. Launch Site

VAFB SLC 4E has no launches above 10 000 kg. - supposedly, implying a site-level constraint. CCAFS SLC 40 and KSC LC 39A handle a wide payload range. Successful landings occur across all mass ranges. We can see that with increases in payload mass the success rate does not seem to decrease, as launches with highest payloads still successfully land (except for the maximum payload at KSC LC 39A, but we cannot tell if the failure was caused by the payload mass or not given the site - while launch with the same payload mass from CCAFS SLC 40 finished in successful landing).



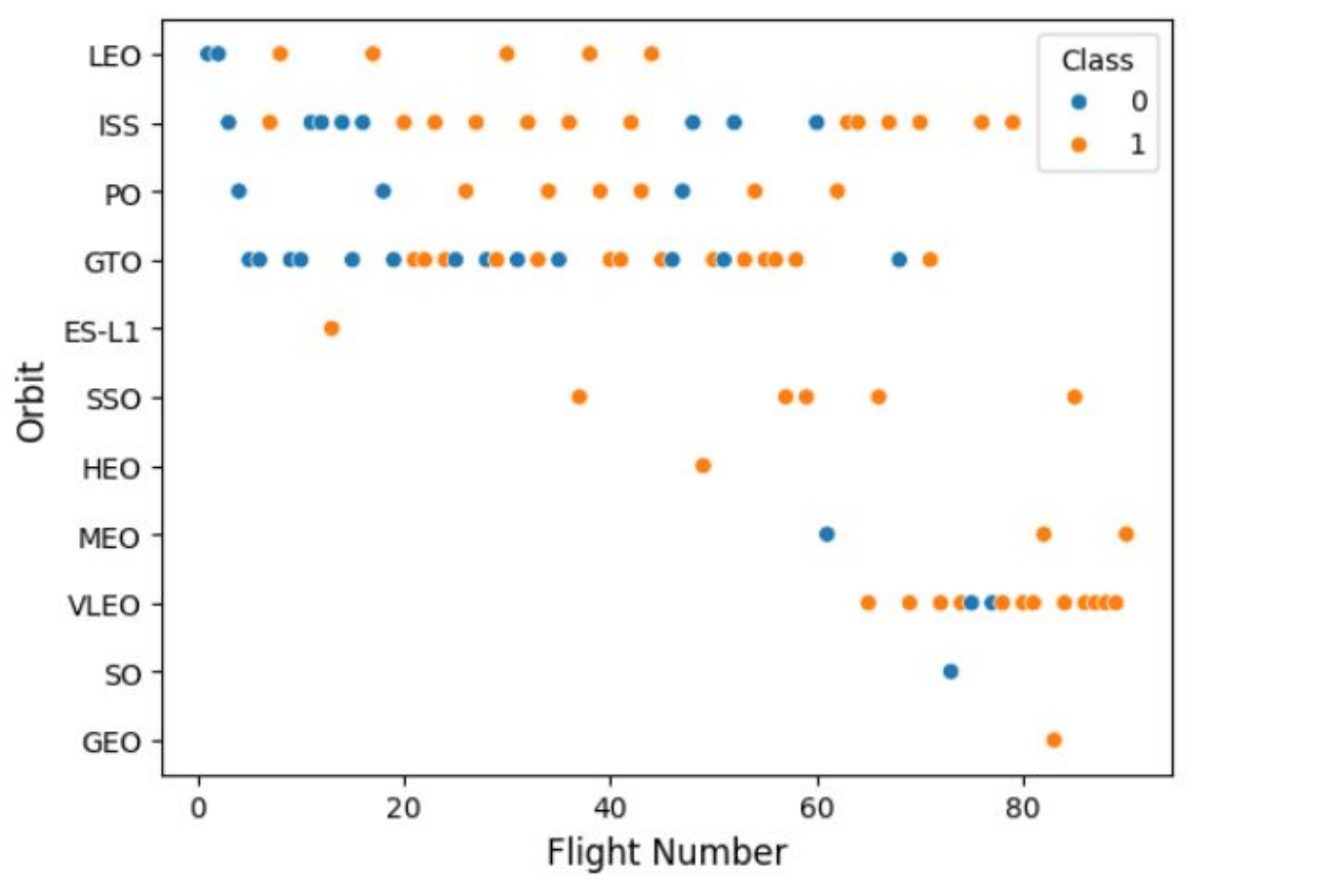
Success Rate vs. Orbit Type



GEO, HEO, SSO, ES-L1 show 100% success. GTO is the most volatile - both successes and failures present. LEO, MEO, PO and ISS average ~60–70%, improving over time. VLEO shows high success. SO seems to either not have any successes or it is an orbit to which there had not been any missions launched.

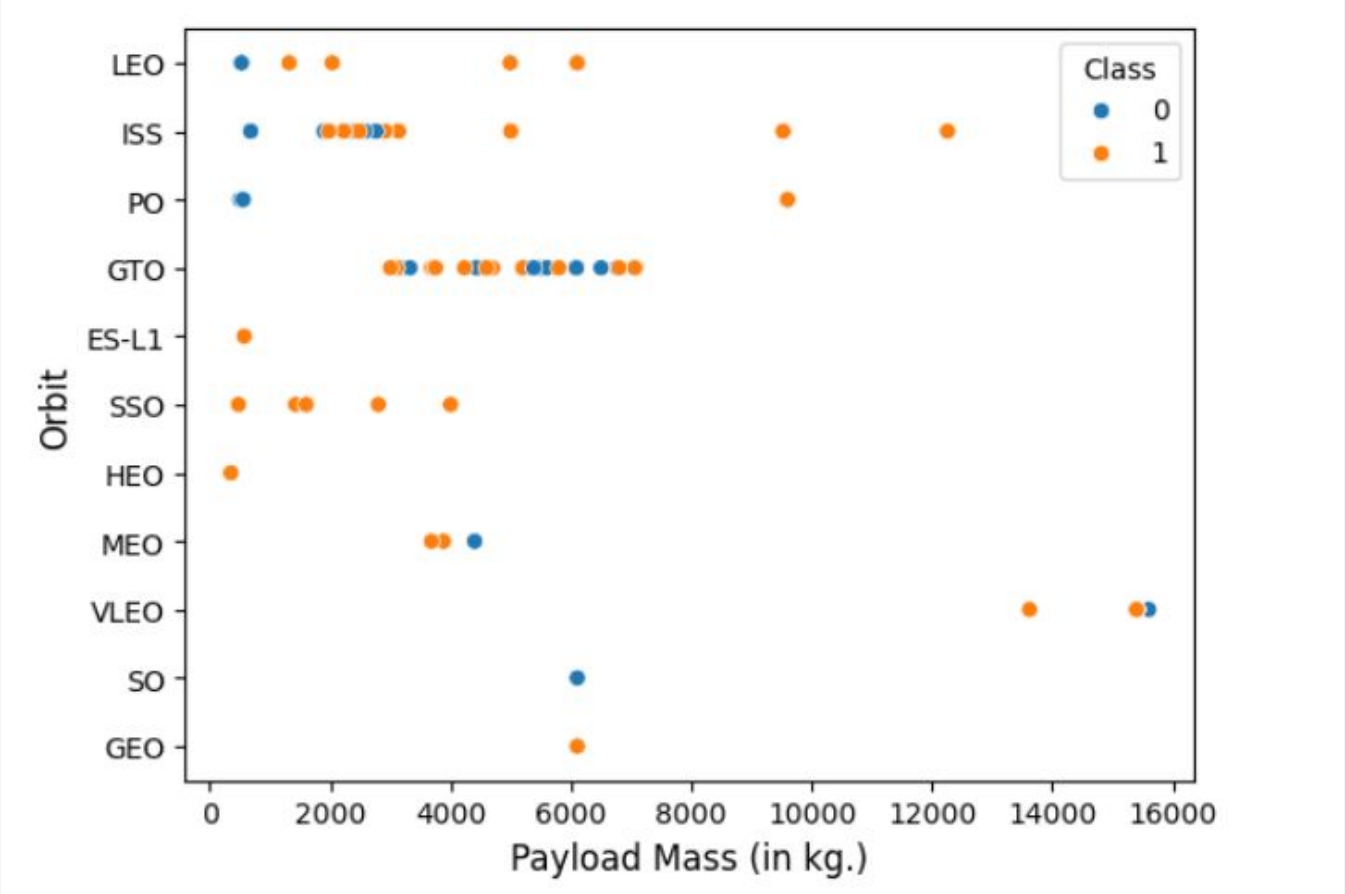
Flight Number vs. Orbit Type

In LEO, a clear trend exists - success improves with flight number. GTO shows no significant dependence on flight number. ISS, PO, MEO and VLEO show consistent success in later launches. All of the launches to SSO were successful in terms of landing. For ES-L1, HEO, GEO only one mission for each was launched, all with successful landings. The only attempt to launch at SO led to unsuccessful landing.



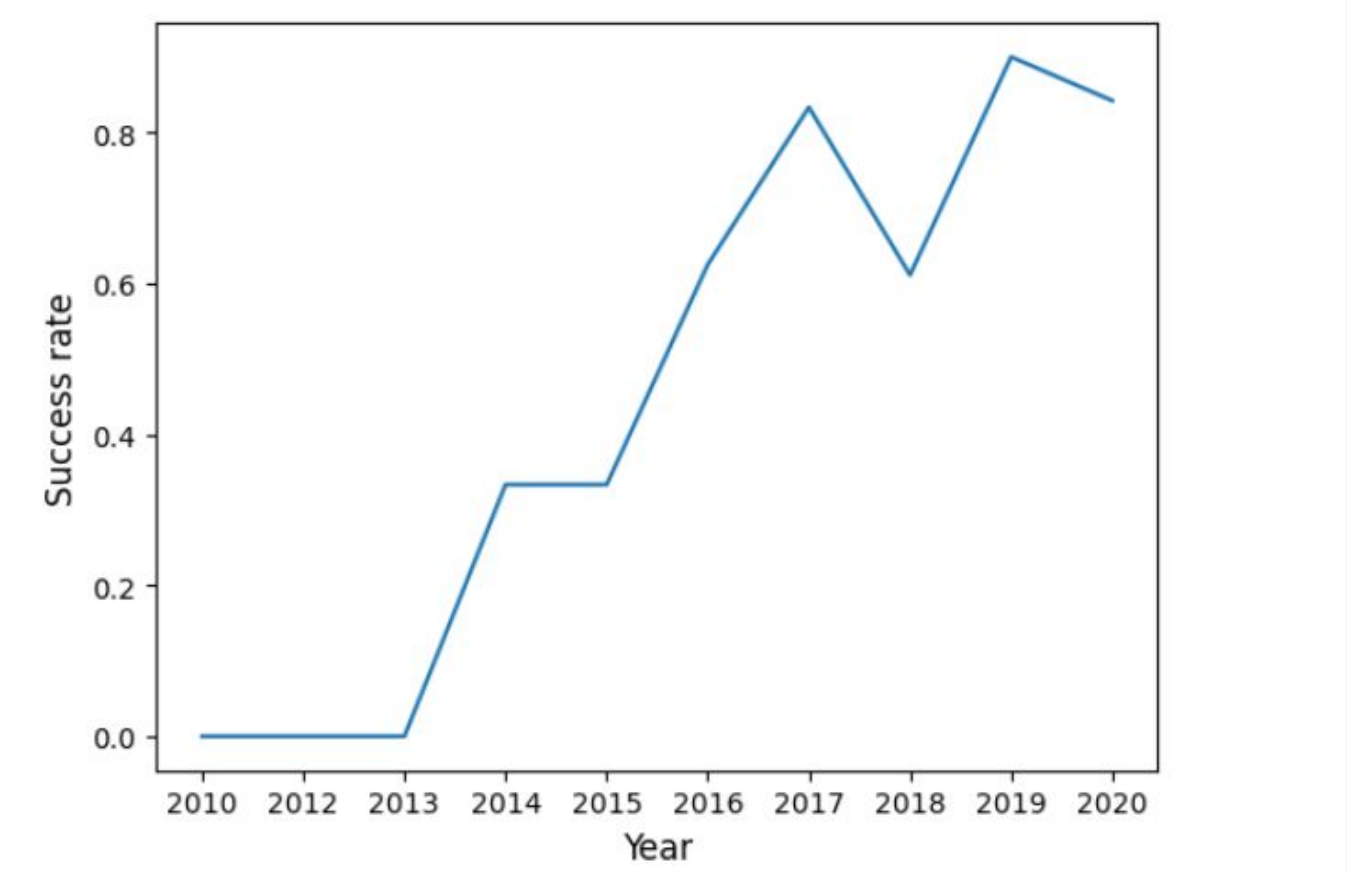
Payload vs. Orbit Type

For PO, LEO and ISS, heavier payloads tend to be more successful. GTO is hard to interpret - both classes appear at any mass. VLEO and SSO succeed across diverse payload ranges, although to be precise mission with highest payload had unsuccessful landing, although second highest, insignificantly different from the first, had success.



Launch Success Yearly Trend

Success rate was near 0 in 2010–2012 (early experiments). A steady upward trend begins in 2013. By 2019–2020, success rate reaches its maximum. This reflects SpaceX's accumulated experience and continued refinement of booster landing technology, although they have their ups and downs.



All Launch Site Names

The query used: **%sql SELECT DISTINCT Launch_Site FROM SPACEXTBL**

Result - 4 unique sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40.

Insight: SpaceX operates on both US coasts: Cape Canaveral in Florida (East Coast) and Vandenberg in California (West Coast).

Launch Site Names Begin with 'CCA'

Query used: **%sql SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5**

Result:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Insights: 5 records from CCAFS LC-40 (the first 5 corresponding to the query). Launches dated 2010–2012, using F9 v1.0 boosters. All 2 early landings failed (Failure parachute), with later ones (taking place in 2012 and 2013) there were no attempts of landing.

Total Payload Mass

Query: **%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)'**

The resulting total payload mass is **45 596 kg.**

Insight: SpaceX has delivered over 45 metric tons to orbit for NASA (CRS), reflecting the scale of the missions.

Average Payload Mass by F9 v1.1

Query: **%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version LIKE 'F9 v1.1%'**

Result: 2 534.7 kg. is the average payload for the F9 v1.1 booster version.

Insight: The result seems to be relatively low compared to later F9 versions, reflecting the limited capacity of this early booster configuration.

First Successful Ground Landing Date

Query: **%sql SELECT Date FROM SPACEXTBL WHERE Landing_Outcome LIKE 'Success (ground pad)' ORDER BY Date LIMIT 1**

Result: the date is **2015-12-22.**

Insight: The first successful ground pad landing occurred on December 22, 2015, becoming a historic milestone for reusable spaceflight.

Successful Drone Ship Landing with Payload between 4000 and 6000

Query: **%sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE Landing_Outcome = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4001 and 5999**

Result: 4 boosters - **F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2**

Insight: Full Thrust (FT) series boosters successfully landed on drone ships with mid-range payloads, confirming the reliability of this landing mode.

Total Number of Successful and Failure Mission Outcomes

Query: **%sql SELECT Mission_Outcome, COUNT(*) AS Total_Number_of_Mission_Outcomes FROM SPACEXTBL GROUP BY Mission_Outcome**

Results:

Failure (in flight) - 1

Success - 99

Success (payload status unclear) - 1

Insight: Great majority of missions succeeded (100/101). Only 1 in-flight failure in the entire launch history.

Boosters Carried Maximum Payload

Query: **%sql SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)**

Results: **F9 B5 B1048.4, F9 B5 B1049.4, F9 B5 B1051.3, F9 B5 B1056.4, F9 B5 B1048.5, F9 B5 B1051.4, F9 B5 B1049.5, F9 B5 B1060.2, F9 B5 B1058.3, F9 B5 B1051.6, F9 B5 B1060.3, F9 B5 B1049.7**

Insight: All record-breaking payloads were carried by Block 5 (B5) boosters - supposedly, the most advanced (and reuse-optimized) version of Falcon 9.

2015 Launch Records

Query:

```
%%sql
   SELECT substr(Date,6,2), Landing_Outcome, Booster_Version, Launch_Site
   FROM SPACEXTBL
   WHERE substr(Date,0,5) = '2015' AND Landing_Outcome = 'Failure (drone ship)'
```

Result:

substr(Date,6,2)	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Insight: Both 2015 drone ship failures used early F9 v1.1 boosters from CCAFS, before the technology was fully mastered.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Query:

```
%%sql
    SELECT Landing_Outcome, COUNT(*)
    FROM SPACEXTBL
    WHERE Date BETWEEN '2010-06-04' and '2017-03-20'
    GROUP BY Landing_Outcome
    ORDER BY COUNT(*) DESC
```

Result:

Landing_Outcome	COUNT(*)
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Insight: During 2010-2017, SpaceX actively experimented with landings - 10 launches without any attempt of landing, 5 drone ship successes and 5 failures, etc. This reflects the company's steep learning curve.

Section 3

Launch Sites Proximities Analysis

All SpaceX Launch Sites - Global View

All 4 launch sites marked using folium.Circle and folium.Marker:

CCAFS LC-40: 28.56°N, 80.58°W (Cape Canaveral, Florida)

CCAFS SLC-40: 28.56°N, 80.58° W

KSC LC-39A: 28.57°N, 80.65°W

VAFB SLC-4E: 34.63°N, 120.61°W (Vandenberg, California)

Observations: All sites are coastal - probably, for safe downrange trajectories over the ocean. The sites are positioned almost as close to the equator as US territory allows, minimizing fuel needed for equatorial orbits.

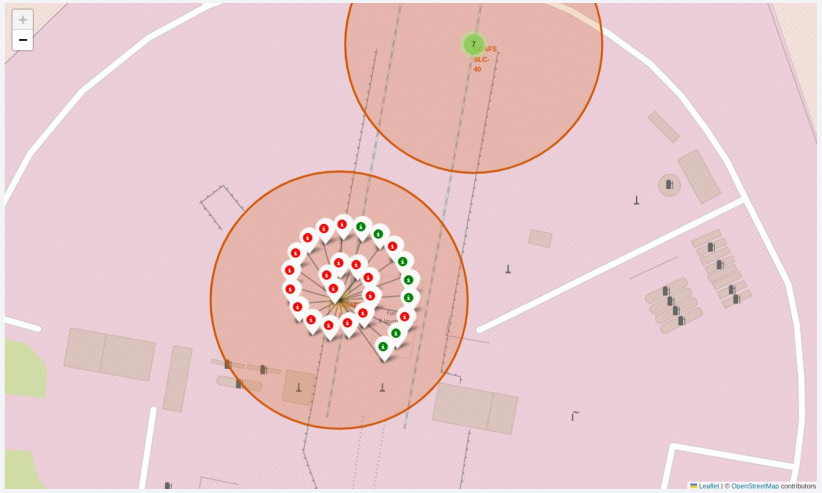
Launch Outcomes by Site - Color-Coded Marker Clusters

MarkerCluster added with color-coded markers for every launch:

Green = successful landing (Class=1)

Red = unsuccessful landing (Class=0)

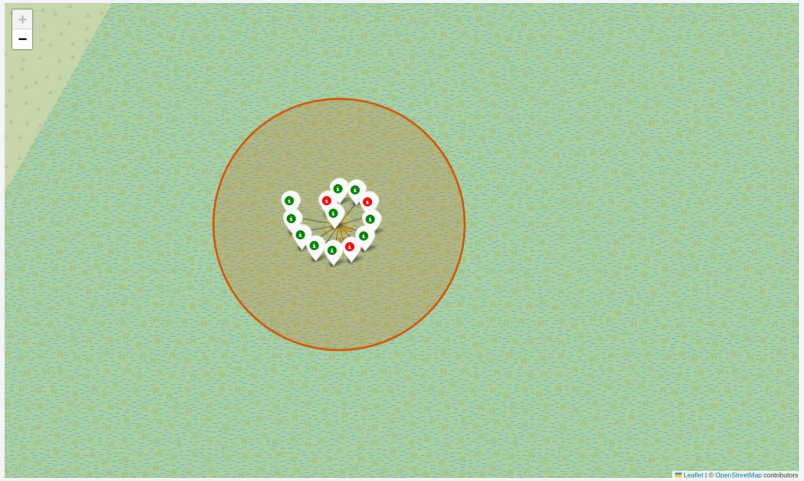
CCAFS LC-40

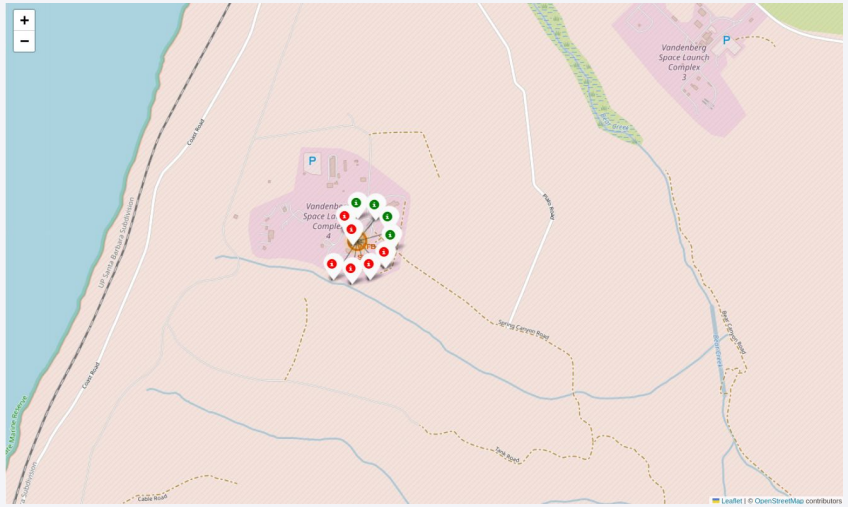


KSC LC-39A

Observations: KSC LC-39A shows the highest concentration of green markers (successes). Early CCAFS launches are predominantly red. Looking at previous data and taking into account timely differences in usage of each site, we could unveil the temporal evolution from failure-heavy early launches to predominant success in later years.

KSC LC-39A



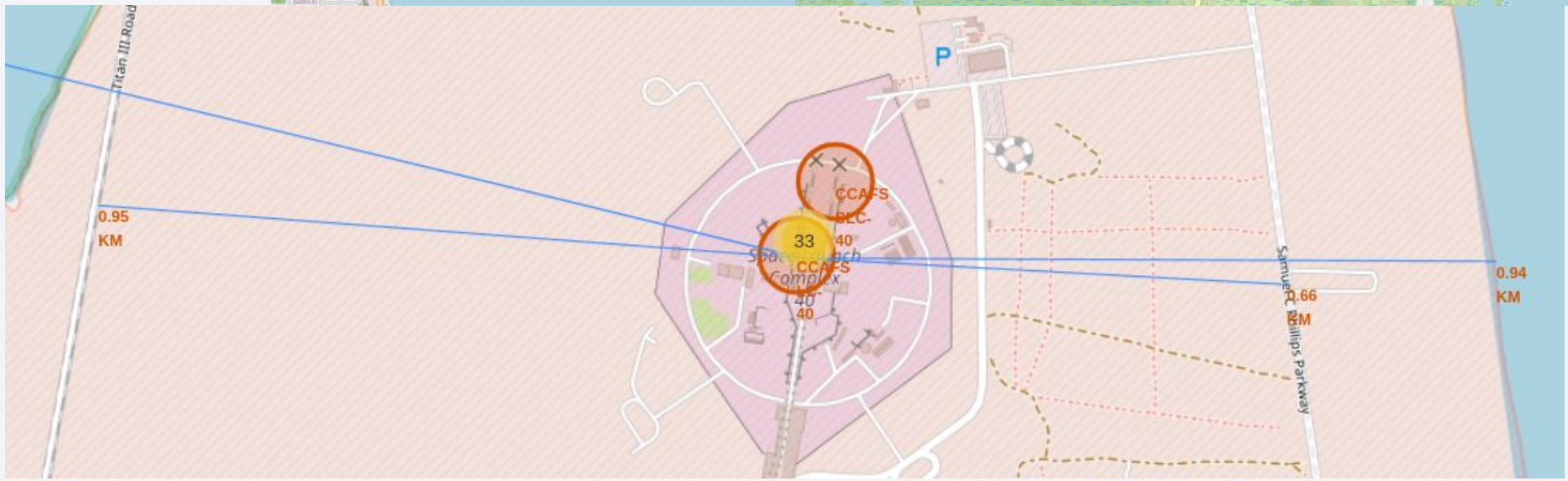


VAFB SLC-4E

Launch Site Proximities — Distance Analysis (CCAFS LC-40)

Using MousePosition and the distance formula, distances from CCAFS LC-40 were calculated:

Feature	Distance
Coastline	~0.96 km
Railway	~0.66 km
Highway	~0.95 km
Nearest city	~23 km

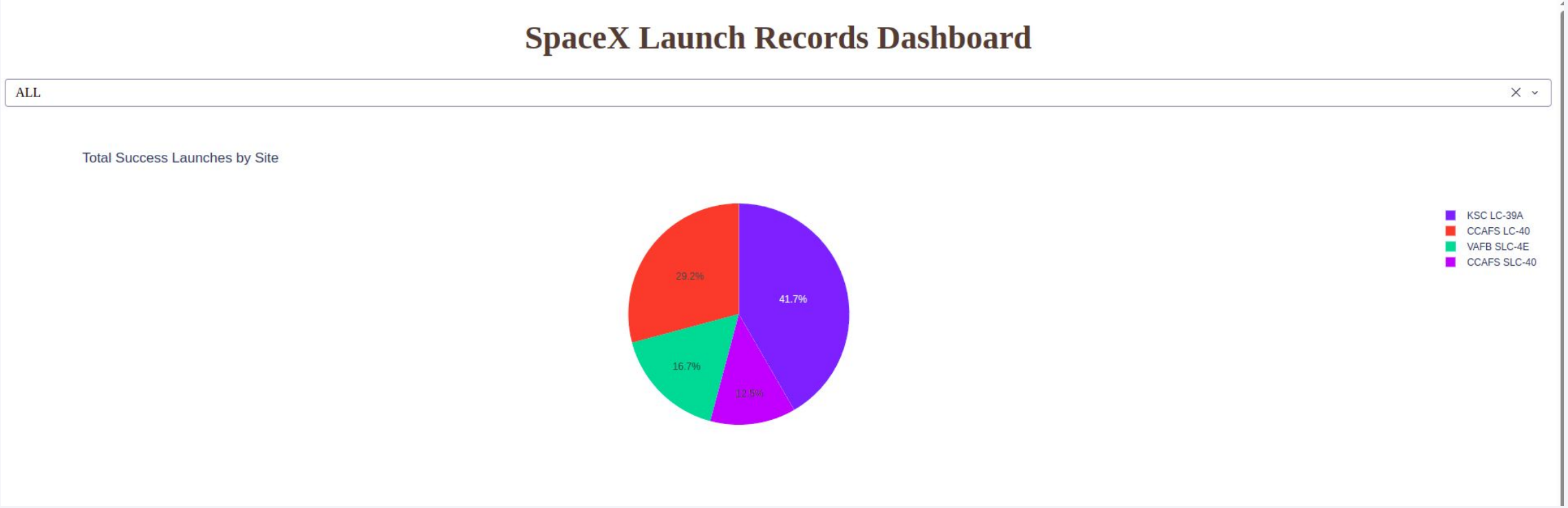


Observations:
The site is immediately adjacent to the coast (probably, for reasons of safety and ocean trajectories), close to transport infrastructure (logistics), but far from populated areas (public safety).

Section 4

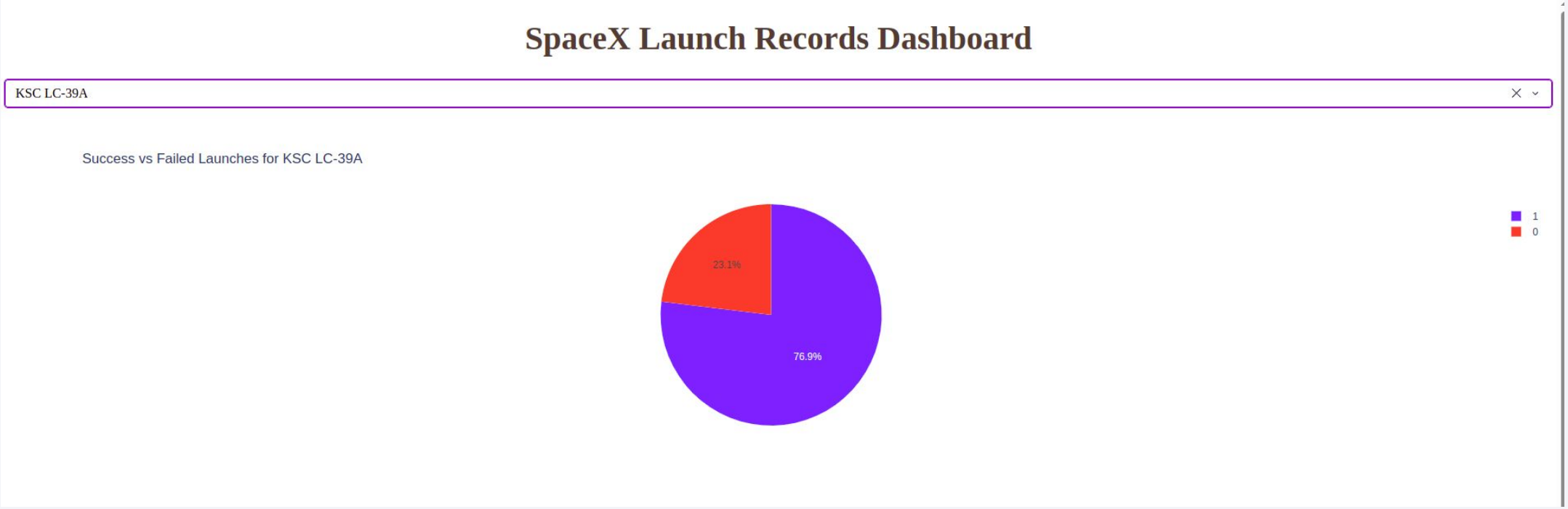
Build a Dashboard with Plotly Dash

Total Launch Success Count by Site - All Sites



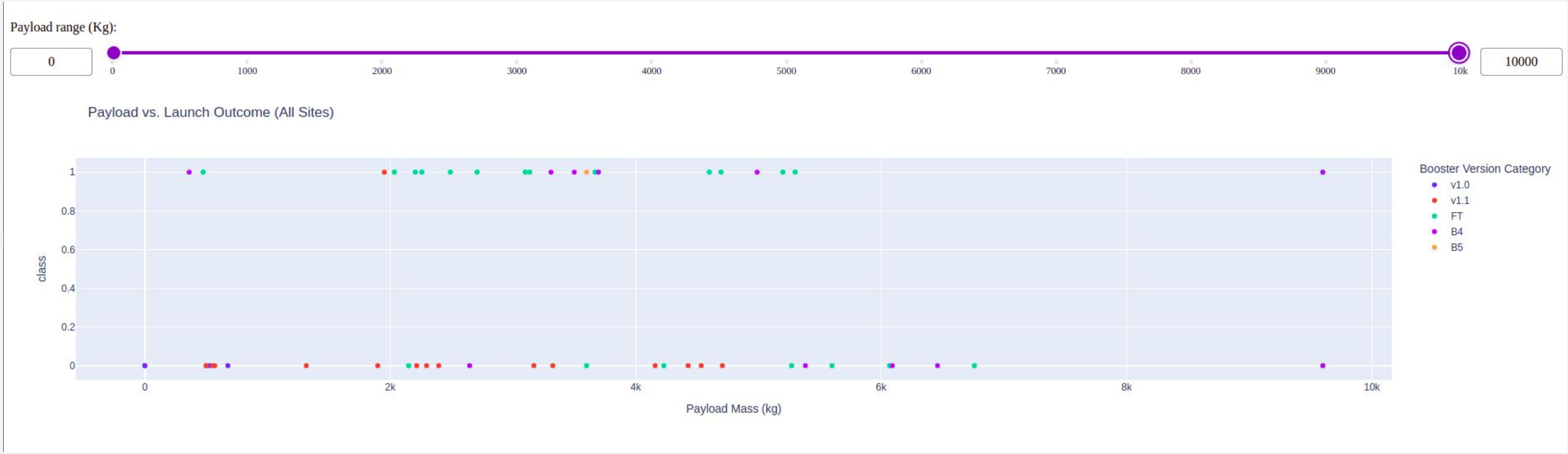
Observations: We can see that most of the success launches were from the site KSC LC-39A, least from CCAFS SLC-40. We have to keep in mind, though, the general amount of launches from each site and the novelty of the site: more novel sites tend to have higher success rate as SpaceX's launches improve over time.

Success vs. Failure - Site with Highest Success Ratio



Observations: Based on the dashboard, KSC LC-39A has the highest success ratio among all sites. As we have found using the internet data, SpaceX received access to this site the last (in 2014), so it probably not the site itself that is so good but rather the results represent the improvement of SpaceX's launches over time.

Payload Mass vs. Launch Outcome - Interactive Scatter Plot



The RangeSlider filters data by payload range (0-10 000 kg., step 1 000).

Observations:

- Heavier payloads (4 000-10 000 kg.) succeed with B4 and FT boosters, but the failure rate is still high - these boosters are supposedly the only ones that SpaceX is considering to be capable of working with such weights.
- Booster versions such as v1.0 and v1.1 mostly fail.
- Only one launch has been made with booster of version B5 with not very high payload mass (around 3 800 kg.) and it led to success.

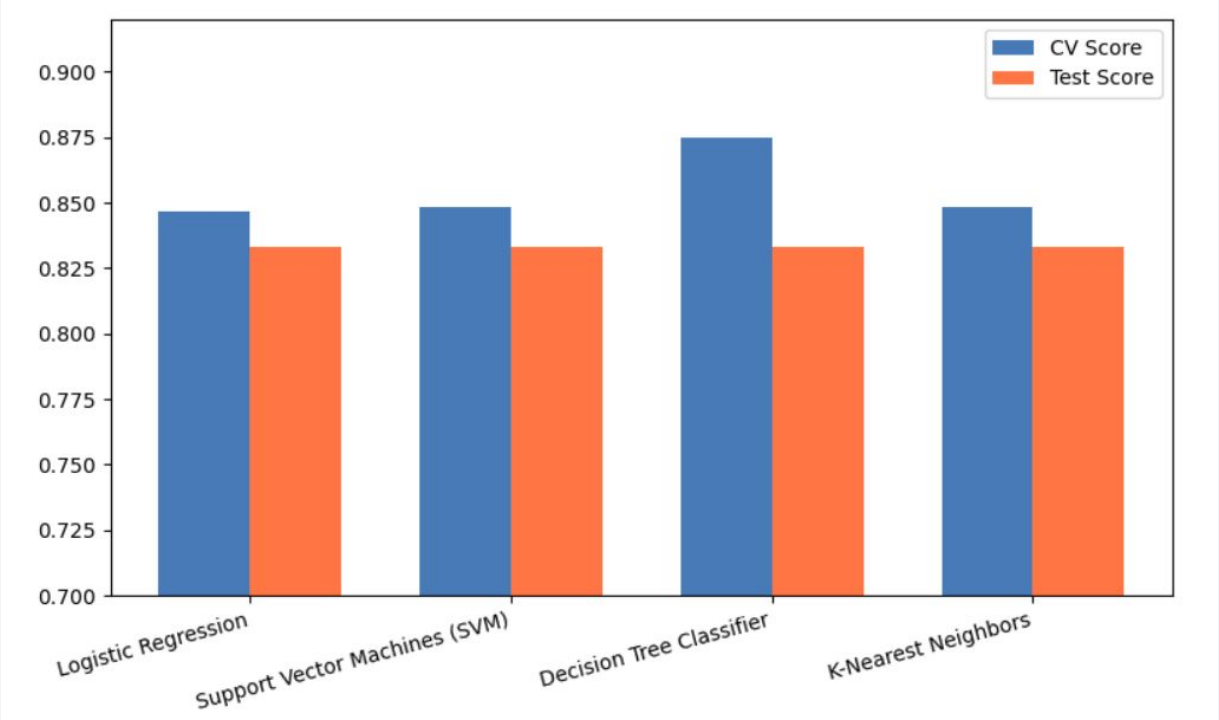
Section 5

Predictive Analysis (Classification)

Classification Accuracy

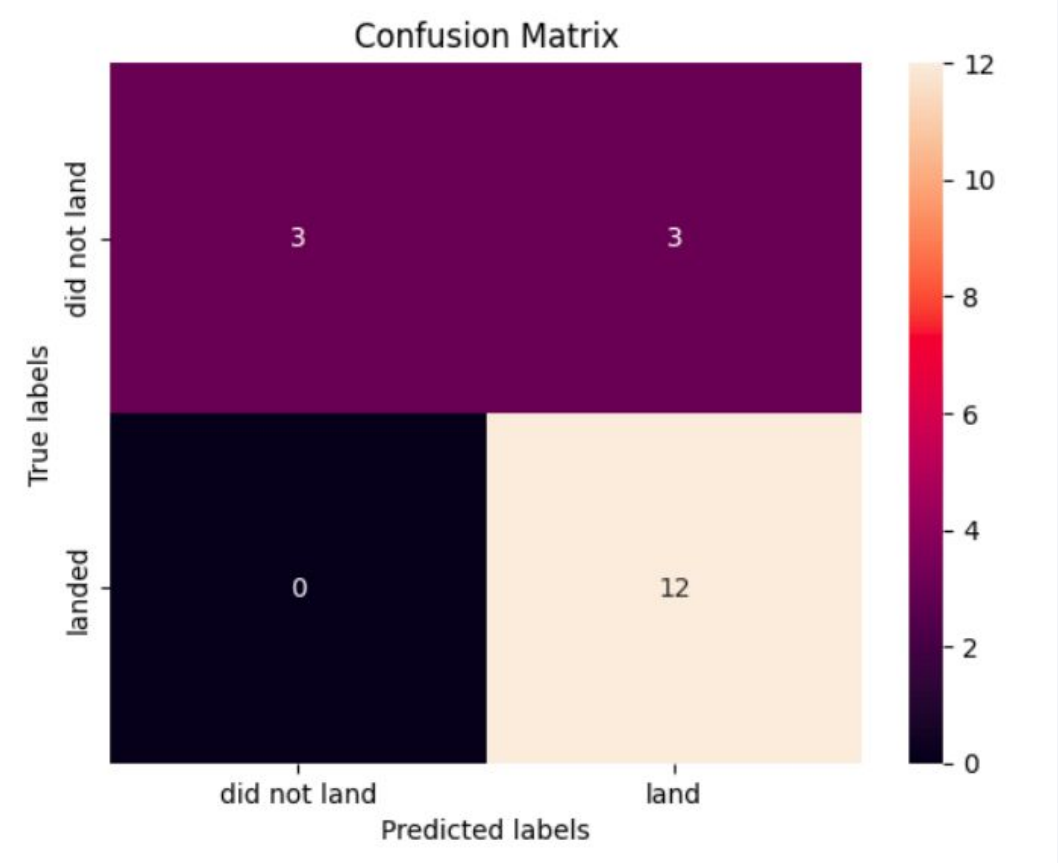
Results: on such a small dataset, and especially such a small test sample, the models perform approximately the same, so in the results of comparison the first one in the list is shown. It is important to either enrich the dataset or at least use the cross-validation score instead of the test score. With cross-validation score, Decision Tree performs the best. Overall best hyperparameters of models:

- LR: C=0.01, penalty=l2, solver=lbfgs
- SVM: C=1.0, gamma=0.032, kernel=sigmoid
- DT: criterion=entropy, max_depth=4, max_features=sqrt, min_samples_leaf=2, min_samples_split=2, splitter=random
- KNN: algorithm=auto, n_neighbors=10, p=1



Confusion Matrix

Results: Out of 18 test samples: 15 correct and 3 errors. The errors are all False Positives: the model predicted a landing when there was none.
Insight: The main issue is false positives (FP=3). This is partly a consequence of the very small test set (18 samples). All 4 models made identical errors on the same examples - a strong indicator that the test set is too small to differentiate model performance.



Conclusions

- Success improves over time: from 2013 to 2020, landing success rate increased steadily, approaching 100% in the most recent years
- Key success factors: flight number (accumulated experience), orbit type (LEO and ISS more reliable than GTO), launch site (KSC LC 39A has the highest success ratio).
- Payload mass is not a limiting factor for B4 and FT boosters - they successfully land with payloads up to 10 000 kg. at least. Maybe B5 boosters can work with even higher payloads, but our task limit was 10 000 kg.
- Geographic advantages of launch sites: coastal proximity, distance from cities, access to transport infrastructure.
- Best ML model: Decision Tree with CV-accuracy of 87.7%; all models achieved 83.3% test accuracy (limited by small test set of 18 samples).
- Practical takeaway: given known launch parameters, the landing outcome can be predicted with ~83–88% accuracy - enabling reliable cost estimation.

Appendix

For model comparison in predictive analysis I have used automated best model search:

```
models_n_scores = {'Logistic Regression': logreg_cv.score(X_test, Y_test),
                   'Support Vector Machines (SVM)': svm_cv.score(X_test, Y_test),
                   'Decision Tree Classifier': tree_cv.score(X_test, Y_test),
                   'K-Nearest Neighbors': knn_cv.score(X_test, Y_test),
                  }

best_model = max(models_n_scores, key=models_n_scores.get)
print(f'Best model: {best_model}, score: {models_n_scores[best_model]}')

models_n_scores_cv = {'Logistic Regression': logreg_cv.best_score_,
                   'Support Vector Machines (SVM)': svm_cv.best_score_,
                   'Decision Tree Classifier': tree_cv.best_score_,
                   'K-Nearest Neighbors': knn_cv.best_score_,
                  }

best_model = max(models_n_scores_cv, key=models_n_scores_cv.get)
print(f'Best model: {best_model}, score: {models_n_scores_cv[best_model]}')
```

To represent distances of the chosen launch site to key objects, I have used a loop:

```
lats_longs = [[28.61091, -80.80618], [28.56291, -80.58701], [28.56194,
-80.57057]]
for lat, long in lats_longs:
    distance = calculate_distance(launch_site_lat, launch_site_lon, lat, long)
    coordinate = [lat, long]
    distance_marker = folium.Marker(
        coordinate,
        icon=DivIcon(
            icon_size=(20,20),
            icon_anchor=(0,0),
            html='<div style="font-size: 12; color:#d35400;"><b>%s</b></div>' %
"{:10.2f} KM".format(distance),
            )
        )
    lines=folium.PolyLine(locations=[coordinate, [launch_site_lat,
launch_site_lon]], weight=1)

    site_map.add_child(distance_marker)
    site_map.add_child(lines)

site_map
```

Thank you!